

Practice Intermediate Algebra Test IV (Version 1)

Name
Date
Course

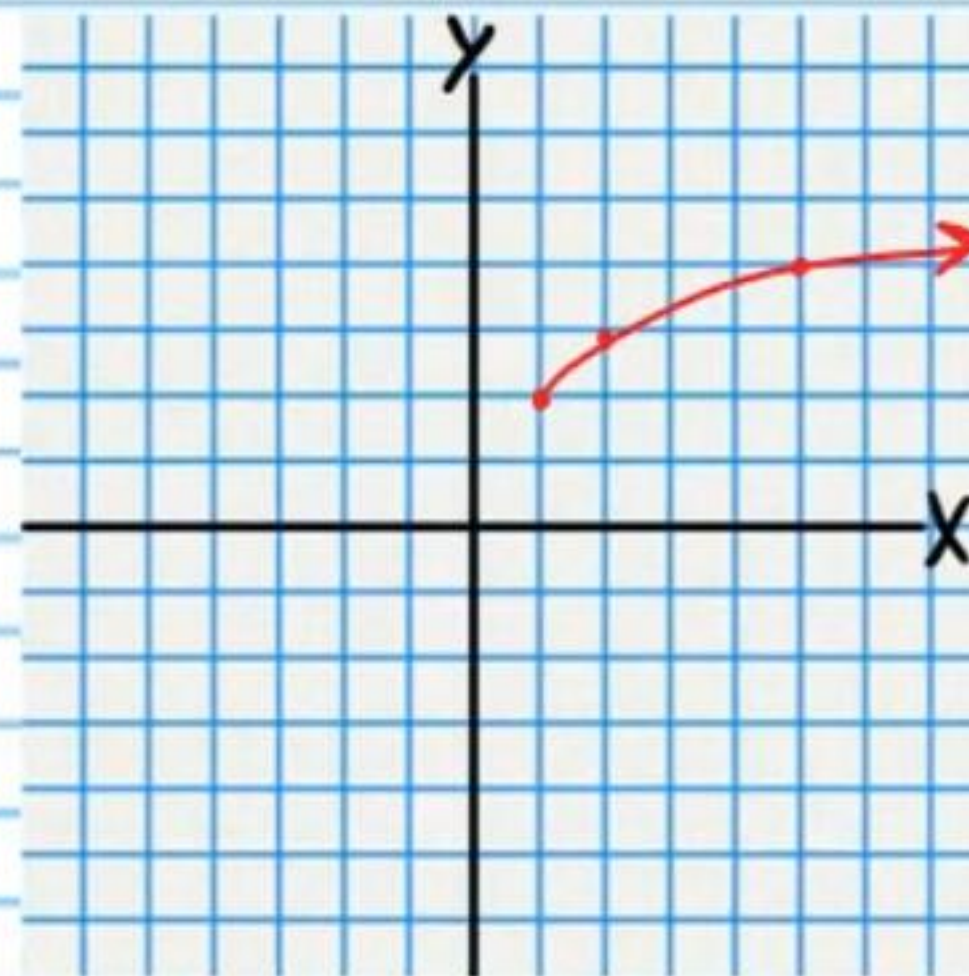
- I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- II) Show all work.
- III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for this test.
- V) Each problem is worth four points for a total of 100 points.

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation.

① $y - 2 = \sqrt{x - 1}$

Parent Relation: Square Root

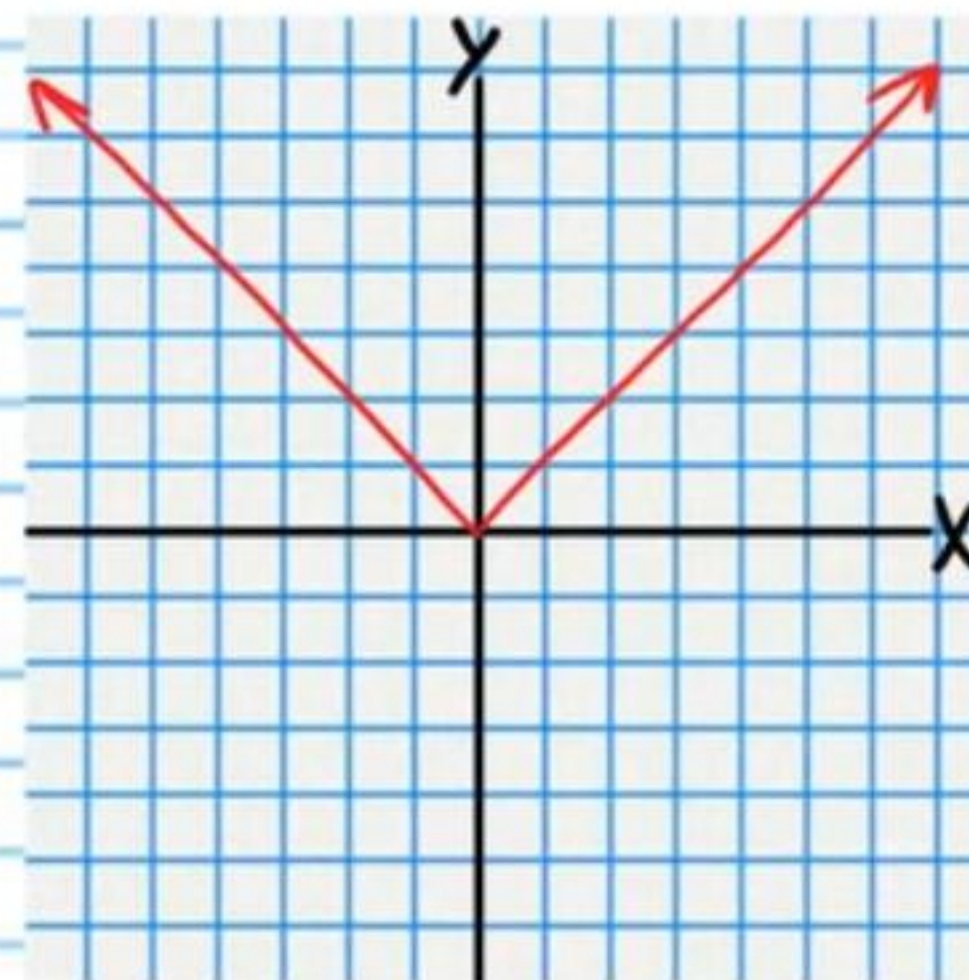
Parent Equation: $y = \sqrt{x}$



② $y = |-x|$

Parent Relation: Absolute Value

Parent Equation: $y = |x|$



Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Draw the asymptote(s) on the graph as dotted lines and write their equation(s). If applicable, identify the type of dilation involved.

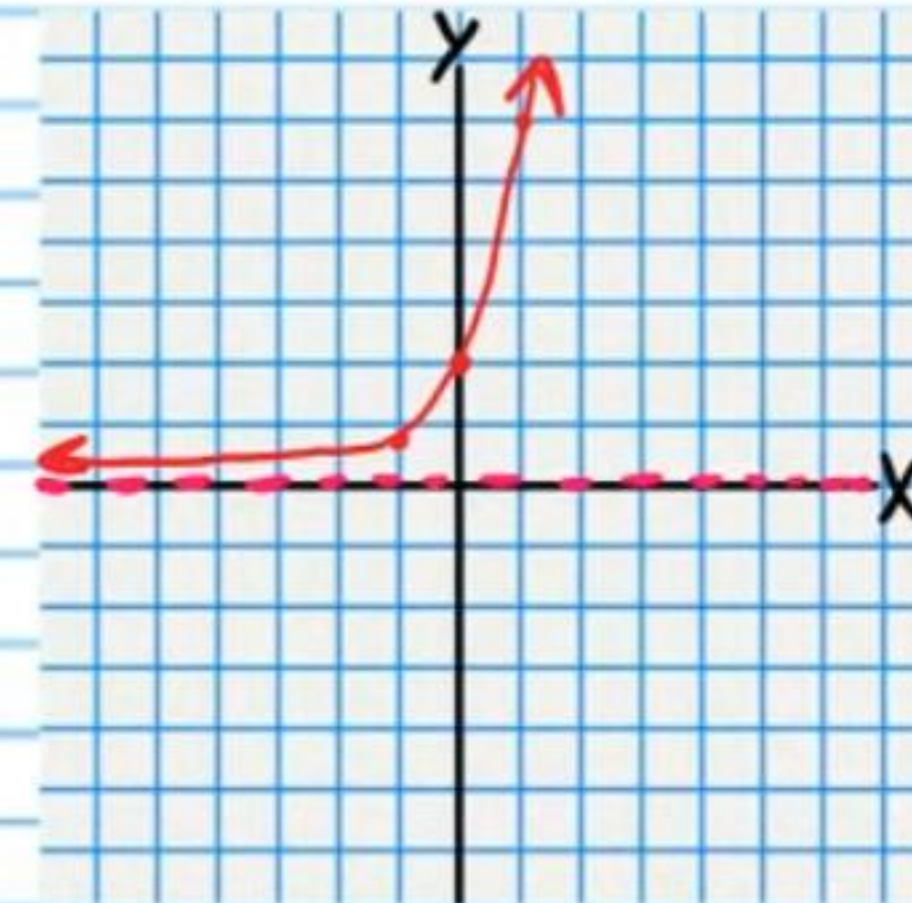
③ $\frac{y}{2} = 3^x$

Parent Relation: Exponential

Parent Equation: $y = 3^x$

Dilation: Vertical Stretch

Asymptote Equation: $y = 0$

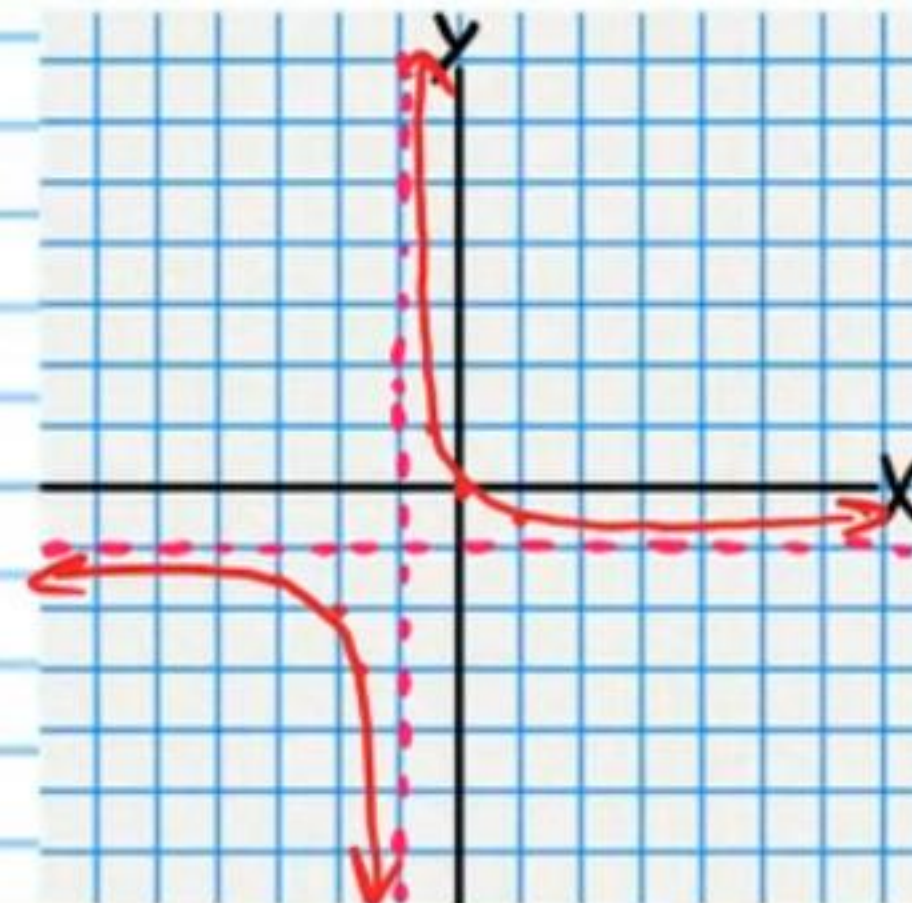


④ $y + 1 = \frac{1}{x + 1}$

Parent Relation: Reciprocal

Parent Equation: $y = \frac{1}{x}$

Asymptote Equations: $x = -1, y = -1$



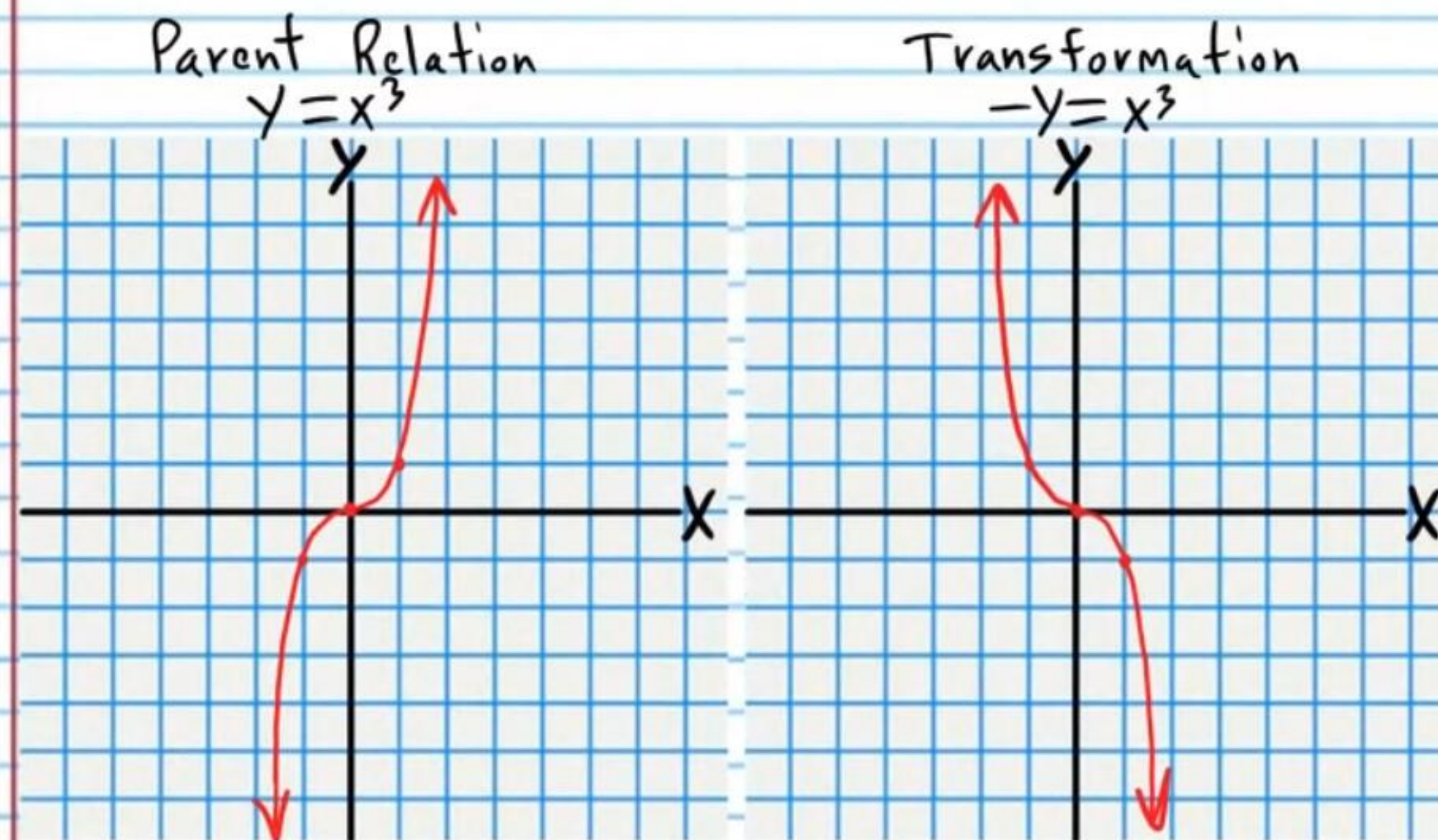
Change the form of each equation below so that it can be graphed with the transformation method. Draw the parent graph. Then draw each transformation, and write the equation for each graph. Draw any asymptotes as dotted lines.

⑤ $y = -x^3$

Change the form of the equation.

$\hookrightarrow -y = x^3$

Altered Form:



⑥ $y = -\log_3 X - 1$

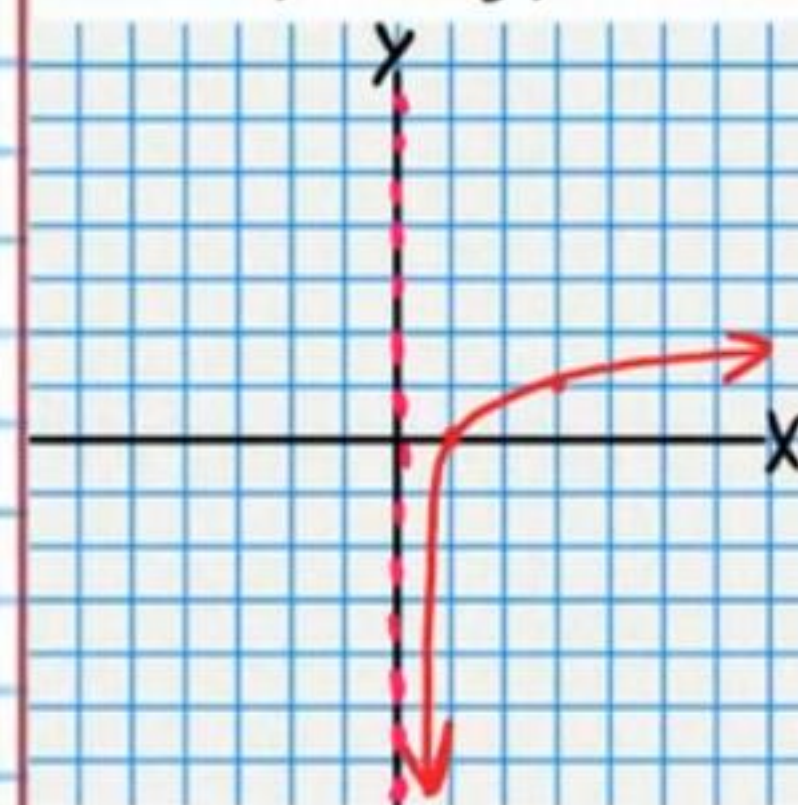
Change the form of the equation.

$\hookrightarrow y + 1 = -\log_3 X$

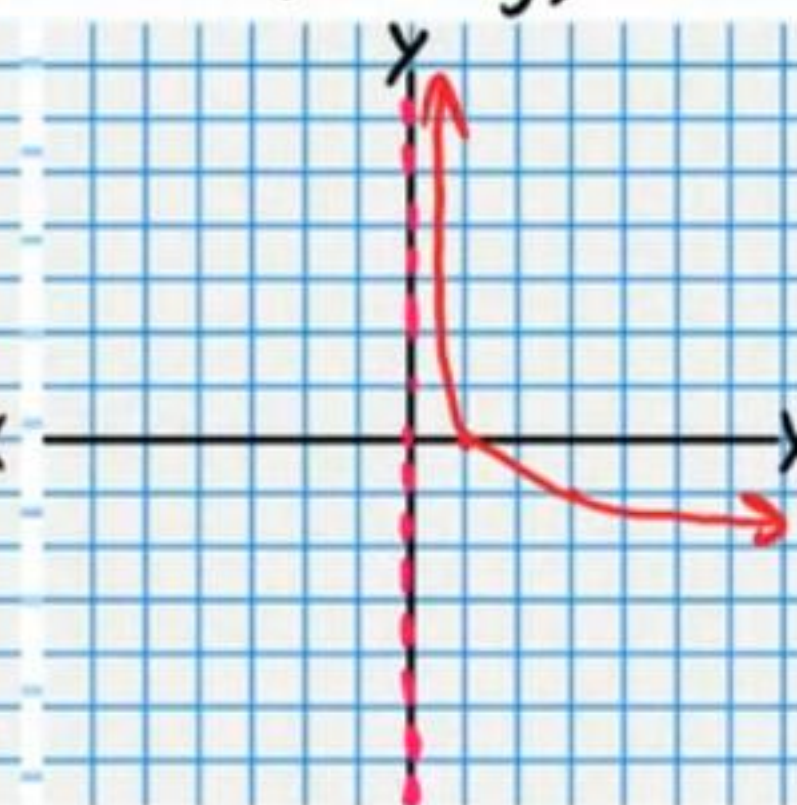
$-(y + 1) = \log_3 X$

Altered Form: $-(y + 1) = \log_3 X$

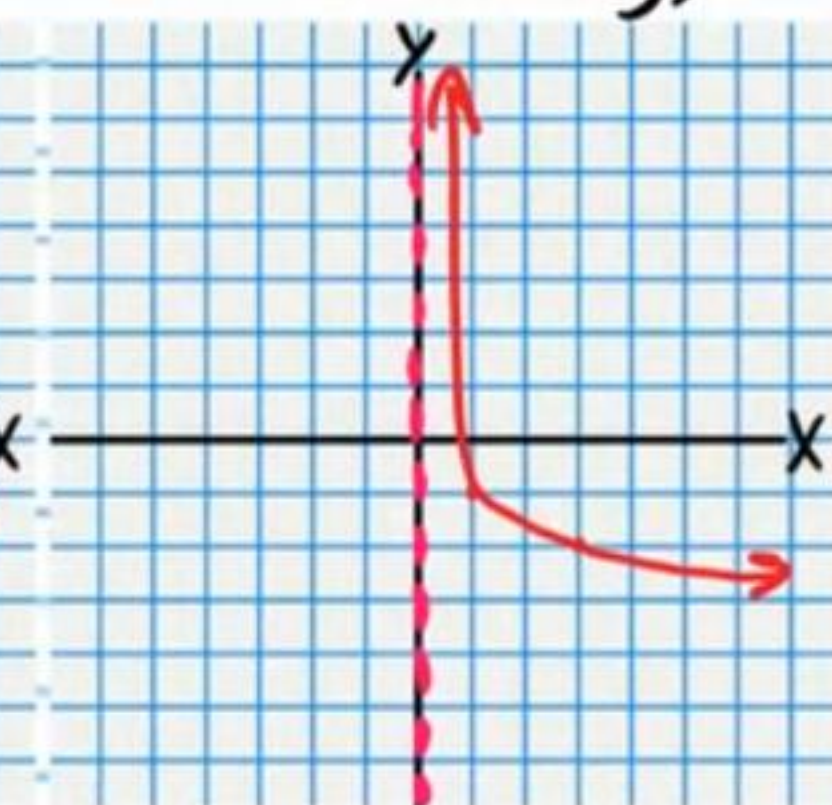
Parent Relation
 $y = \log_3 X$



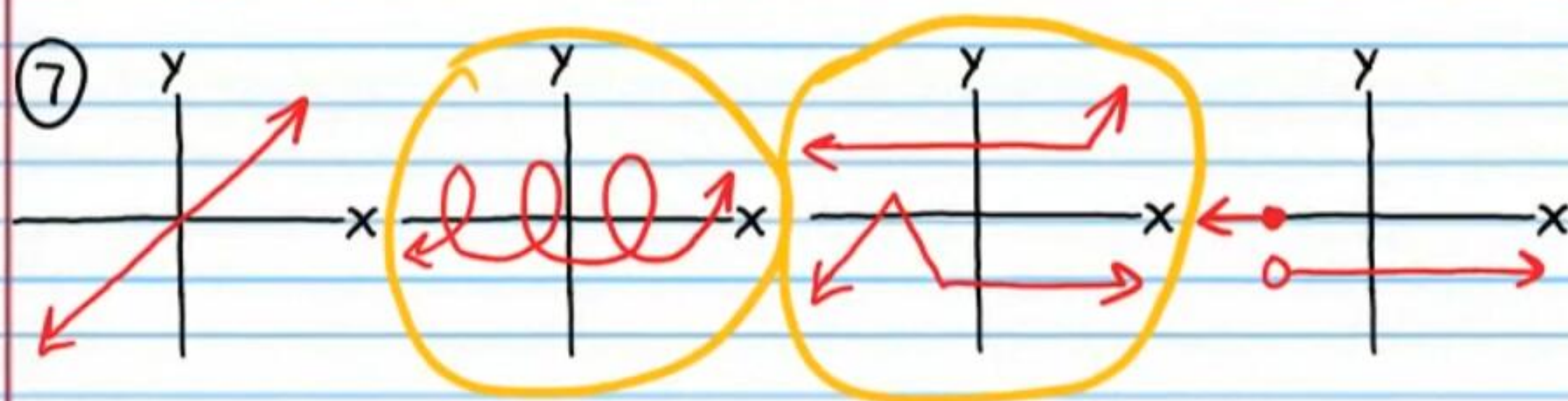
Transformation #1
 $-y = \log_3 X$



Transformation #2
 $-(y + 1) = \log_3 X$



Circle the relation(s) below that cannot be functions.



⑧ If $f(x) = \frac{|x|}{3} + 5$, fill in the table below.

x	-3	0	π	6
f(x)	6	5	$\frac{\pi}{3} + 5$	7

i) $f(-3) = \frac{|-3|}{3} + 5 = \frac{3}{3} + 5 = 1 + 5 = \boxed{6}$

ii) $f(0) = \frac{|0|}{3} + 5 = \frac{0}{3} + 5 = 0 + 5 = \boxed{5}$

iii) $f(\pi) = \frac{|\pi|}{3} + 5 = \boxed{\frac{\pi}{3} + 5}$

iv) $f(6) = \frac{|6|}{3} + 5 = \frac{6}{3} + 5 = 2 + 5 = \boxed{7}$

⑨ Given the function below, find the values.

$$f(x) = \begin{cases} x+1, & 0 < x < 10 \\ 3, & 10 \leq x < 20 \end{cases}$$

i) $f(-1) = \boxed{\text{undefined}}$

iii) $f(10) = \boxed{3}$

ii) $f(8) = 8 + 1 = \boxed{9}$

iv) $f(36) = \boxed{\text{undefined}}$

⑩ If $f(x) = x^2$ and $g(x) = \frac{1}{x}$, find the following:

i) $(f+g)(x) = f(x) + g(x) = \boxed{x^2 + \frac{1}{x}}$

ii) $\frac{f}{g}(-2) = \frac{f(-2)}{g(-2)} = \frac{(-2)^2}{\frac{1}{-2}} = \frac{4}{-\frac{1}{2}} = \boxed{-8}$

⑪ Write the domain of the function below in word form.

$$g(x) = \frac{1}{x+2} \quad \begin{matrix} x+2 \neq 0 \\ x \neq -2 \end{matrix}$$

All real numbers except -2.

⑫ Write the domain of the function below in interval notation.

$$h(x) = \sqrt{\frac{x}{2} + 1} \quad \frac{x}{2} + 1 \geq 0$$

$$\frac{x}{2} \geq -1$$

$$x \geq -2$$

$$\boxed{[-2, \infty)}$$

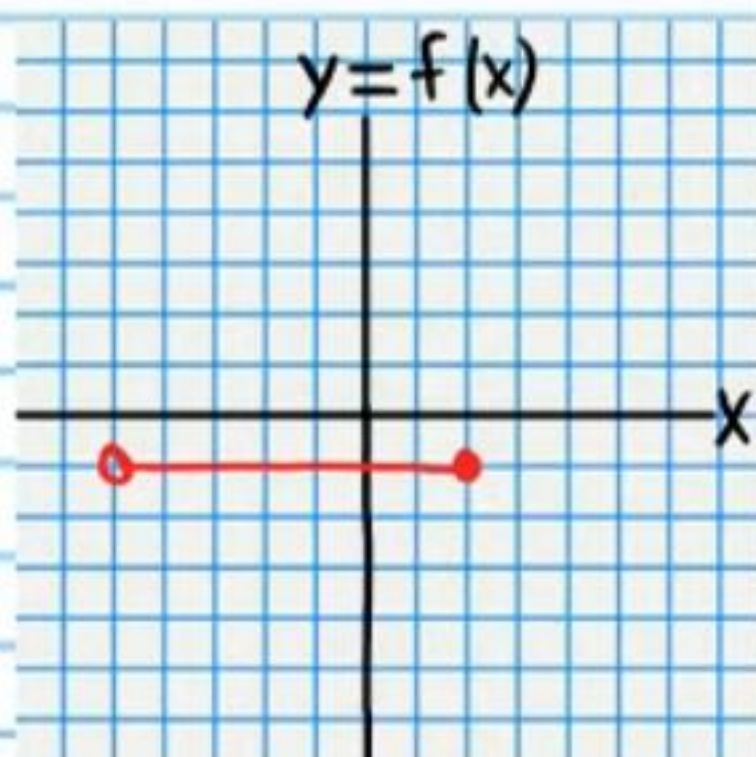
- ⑬ The graph of a function is written below. Write the domain and range of the function in set notation.

Domain

$$\{x | -5 < x \leq 2\}$$

Range

$$\{y | y = -1\}$$

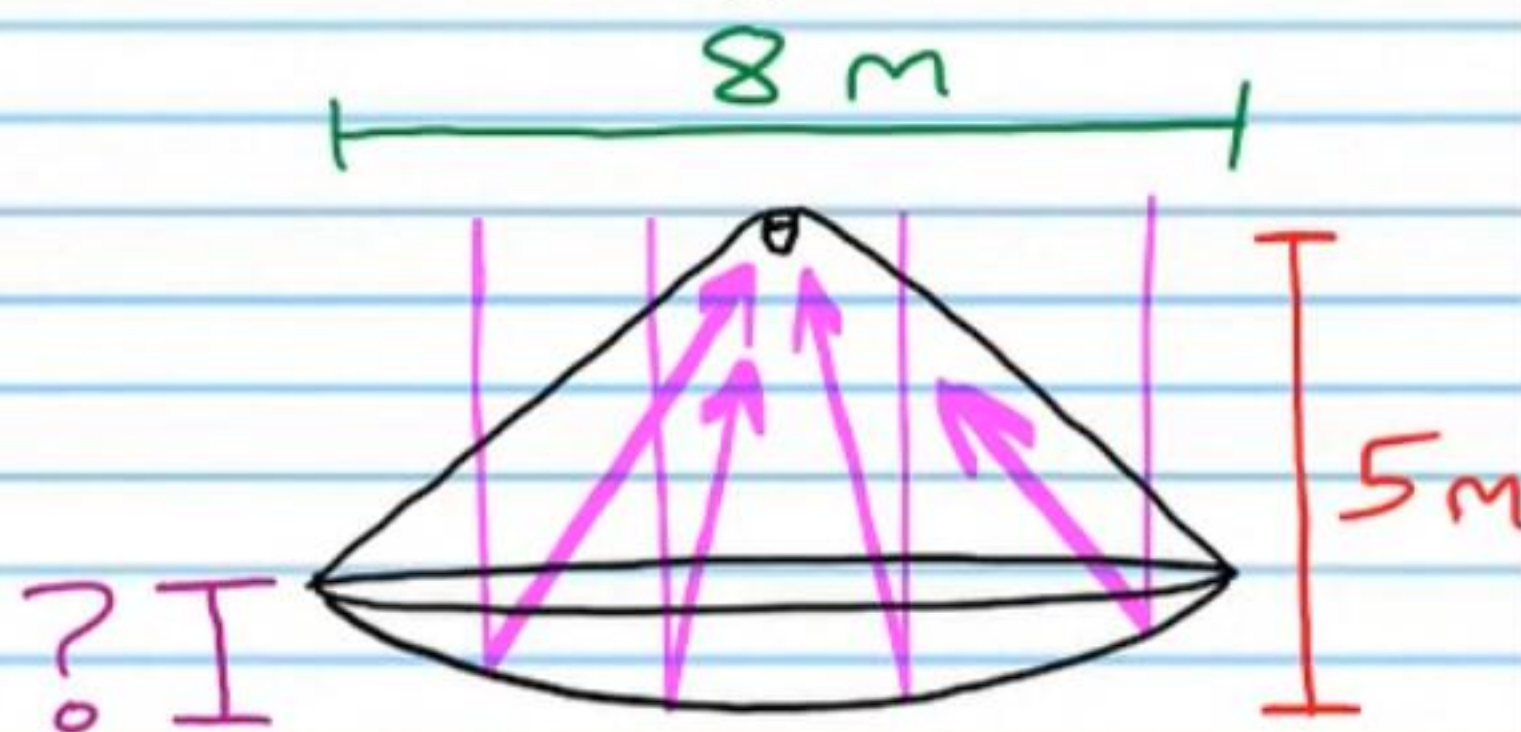


- ⑭ A satellite dish is shaped like a paraboloid. Signals from a satellite hit the dish and are reflected to a single point where a receiver is located. If the receiver is 5 meters from the vertex of the dish, and the opening of the dish is 8 meters wide, how long (i.e. deep) should the dish be?

i) $P = \frac{1}{4a}$

$$5 = \frac{1}{4a}$$

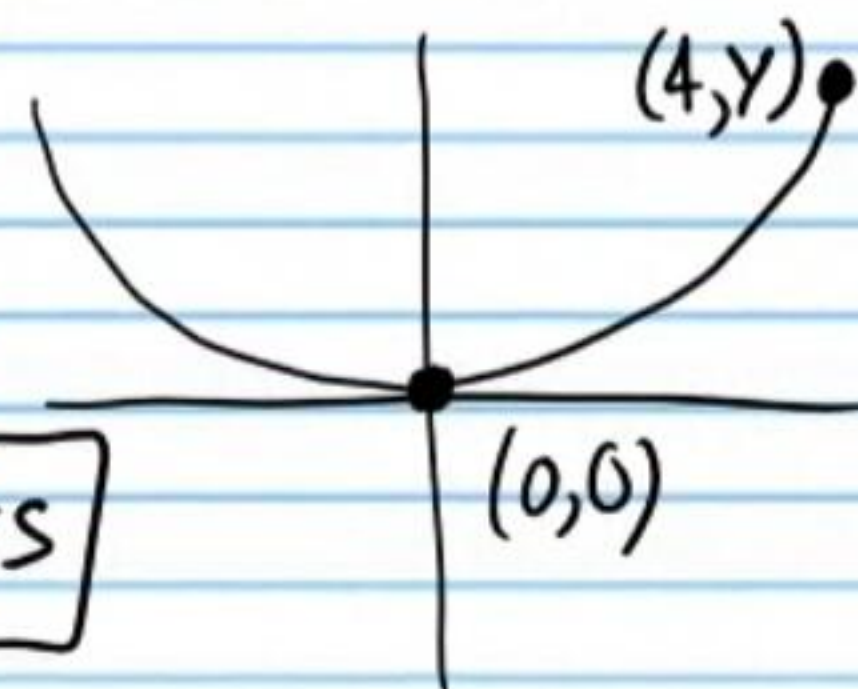
$$a = \frac{1}{20}$$



ii) $y = a(x-h)^2 + k$

$$y = \frac{1}{20}(4-0)^2 + 0$$

$$y = \frac{1}{20}(16) = \frac{16}{20} = \frac{4}{5} \text{ meters}$$



- ⑮ If an investment grows from \$97,000 to \$140,100 at a 2% annual interest rate compounded continuously, how much time passed during this growth?

$$A = Pe^{rt}$$

$$140,100 = 97,000e^{0.02t}$$

$$1.4443299 = e^{0.02t}$$

$$0.02t \approx \log_e 1.4443299$$

$$0.02t \approx \ln 1.4443299$$

$$0.02t \approx 0.367645$$

$$t \approx 18.382 \text{ years}$$

- ⑯ If \$8,000 is deposited in a bank account paying a 5% annual interest rate compounded monthly, how much money will be in the account in 4 years? Assume that there are no withdrawals or additional deposits.

$$\begin{aligned} A &= P\left(1 + \frac{r}{n}\right)^{nt} = 8,000\left(1 + \frac{0.05}{12}\right)^{12(4)} = 8,000(1.00416)^{48} \\ &= 8,000(1.00416)^{48} \\ &\approx 8,000(1.220895) \\ &\approx \$9,767.16 \end{aligned}$$

Solve. Round to the nearest thousandth.

$$\textcircled{17} -4e^x + 6 = -18$$

$$\textcircled{18} 5\log_2 - \log X = 1$$

$$-4e^x = -24$$

$$5\log_{10} 2 - \log_{10} X = 1$$

$$e^x = 6$$

$$\log_{10} 2^5 - \log_{10} X = 1$$

$$x = \log_e 6$$

$$\log_{10} 32 - \log_{10} X = 1$$

$$x = \ln 6$$

$$\boxed{x \approx 1.792}$$

$$\log_{10} \left(\frac{32}{X} \right) = 1$$

$$10^1 = \frac{32}{X}$$

$$10 = \frac{32}{X}$$

$$10X = 32$$

$$\boxed{X = 3.2}$$

$\textcircled{19}$ The following number is the length of a disturbance on a seismograph 100 Kilometers from the epicenter of the earthquake that caused the disturbance. Find the magnitude of the earthquake.

0.8 millimeters

$$M = \log(1000x) = \log_{10}(1000(0.8)) = \log_{10} 800 \approx \boxed{2.9}$$

$\textcircled{20}$ Write the domain of the function below in set notation. You may use the grids below if necessary.

$\boxed{\text{All real numbers}}$

$\textcircled{21}$ Write the range of the function below in interval notation. You may use the grids below if necessary.

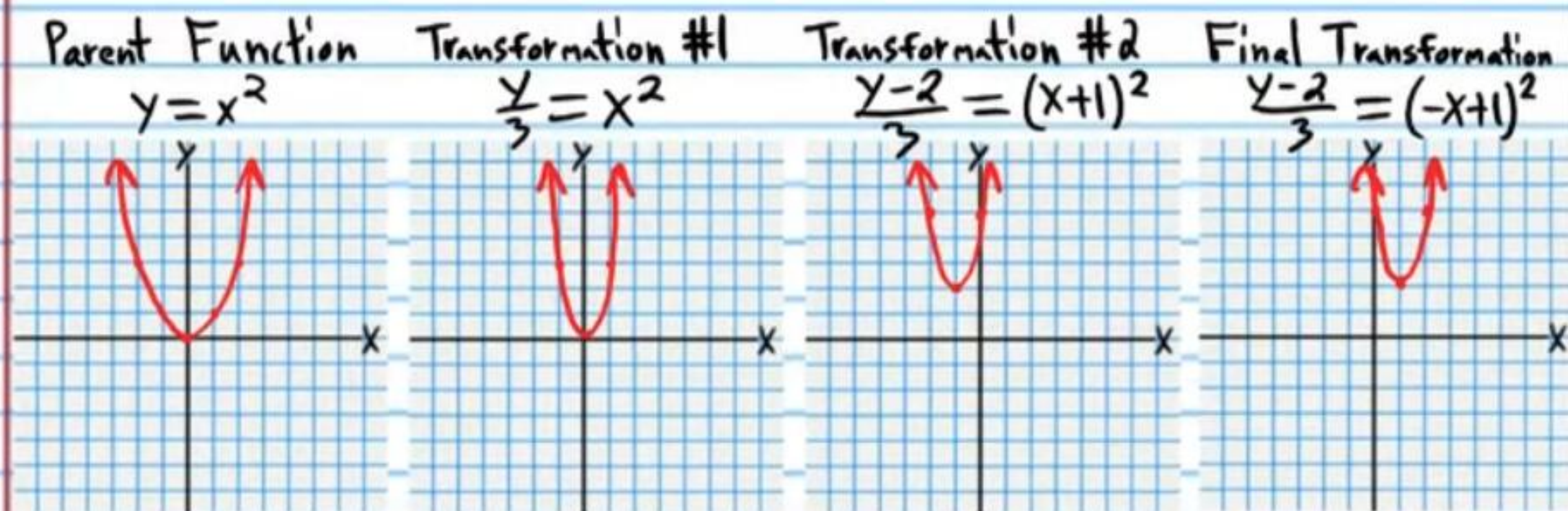
$\boxed{[2, \infty)}$

$$f(x) = 3(-x+1)^2 + 2$$

$$y = 3(-x+1)^2 + 2$$

$$y - 2 = 3(-x+1)^2$$

$$\frac{y-2}{3} = (-x+1)^2$$

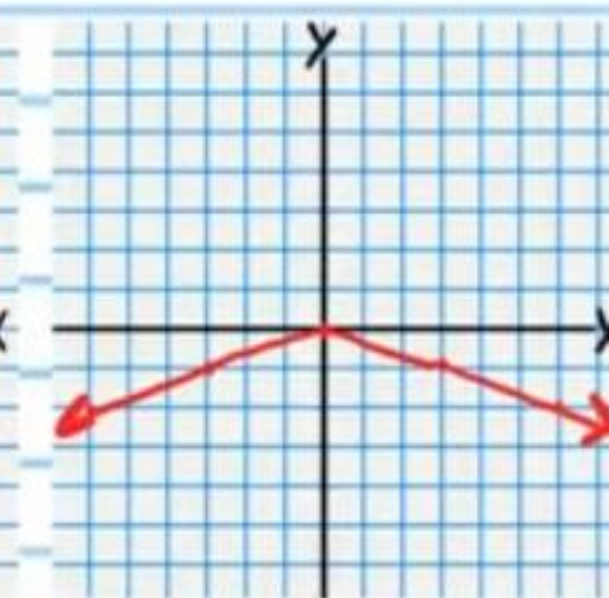
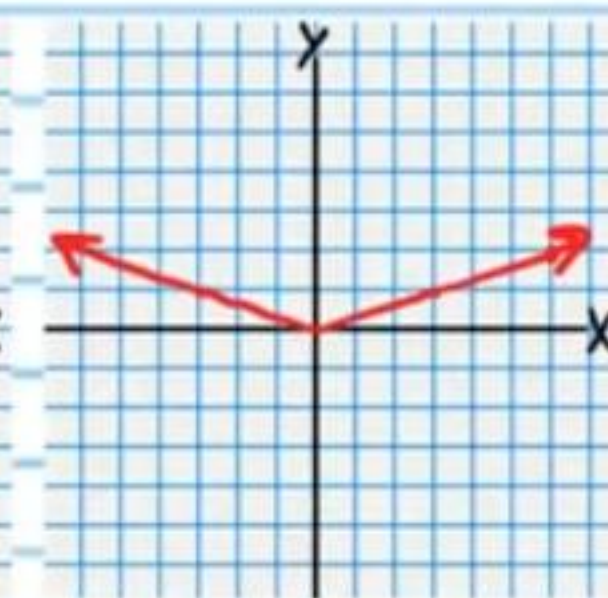
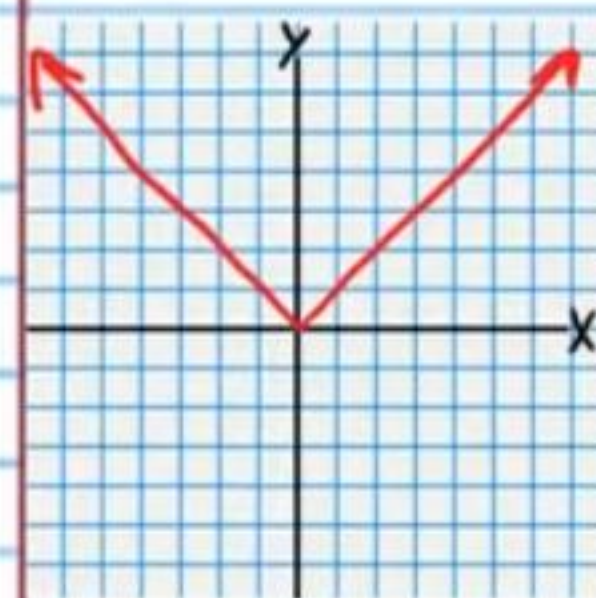


The transformation properties can be used with each pathway at the right to graph the corresponding equation. But in most of these pathways, expressions must be converted into a different form in order to use the transformation properties in the way that they are written. Choose the pathway(s) in which no modification of expressions is required. Then, use your chosen pathway to graph the corresponding equation.

② $-3(y-2) = |x+3|$

Pathway #1	Pathway #2	Pathway #3
$y = x $ ✓	$y = x $ ✓	$y = x $ ✓
$y-2 = x+3 $ ✓	$y-2 = x+3 $ ✓	$3y = x $ ✓
$3(y-2) = x+3 $ ✗	$-(y-2) = x+3 $ ✗	$3(-y) = x $ ✓
$-3(y-2) = x+3 $	$-3(y-2) = x+3 $	$-3(y-2) = x+3 $ ✓
Pathway #4	Pathway #5	Pathway #6
$y = x $	$y = x $	$y = x $
$3y = x $	$-y = x $	$-y = x $
$3(y-2) = x+3 $	$-3y = x $	$-(y-2) = x+3 $
$-3(y-2) = x+3 $	$-3(y-2) = x+3 $	$-3(y-2) = x+3 $

Parent Relation	Transformation #1	Transformation #2	Final Transformation
$y = x $	$3y = x $	$-3y = x $	$-3(y-2) = x+3 $



23 Write the parent relation and the parent equation of the equation below. Then, graph the PARENT relation.

Parent Relation: Reciprocal

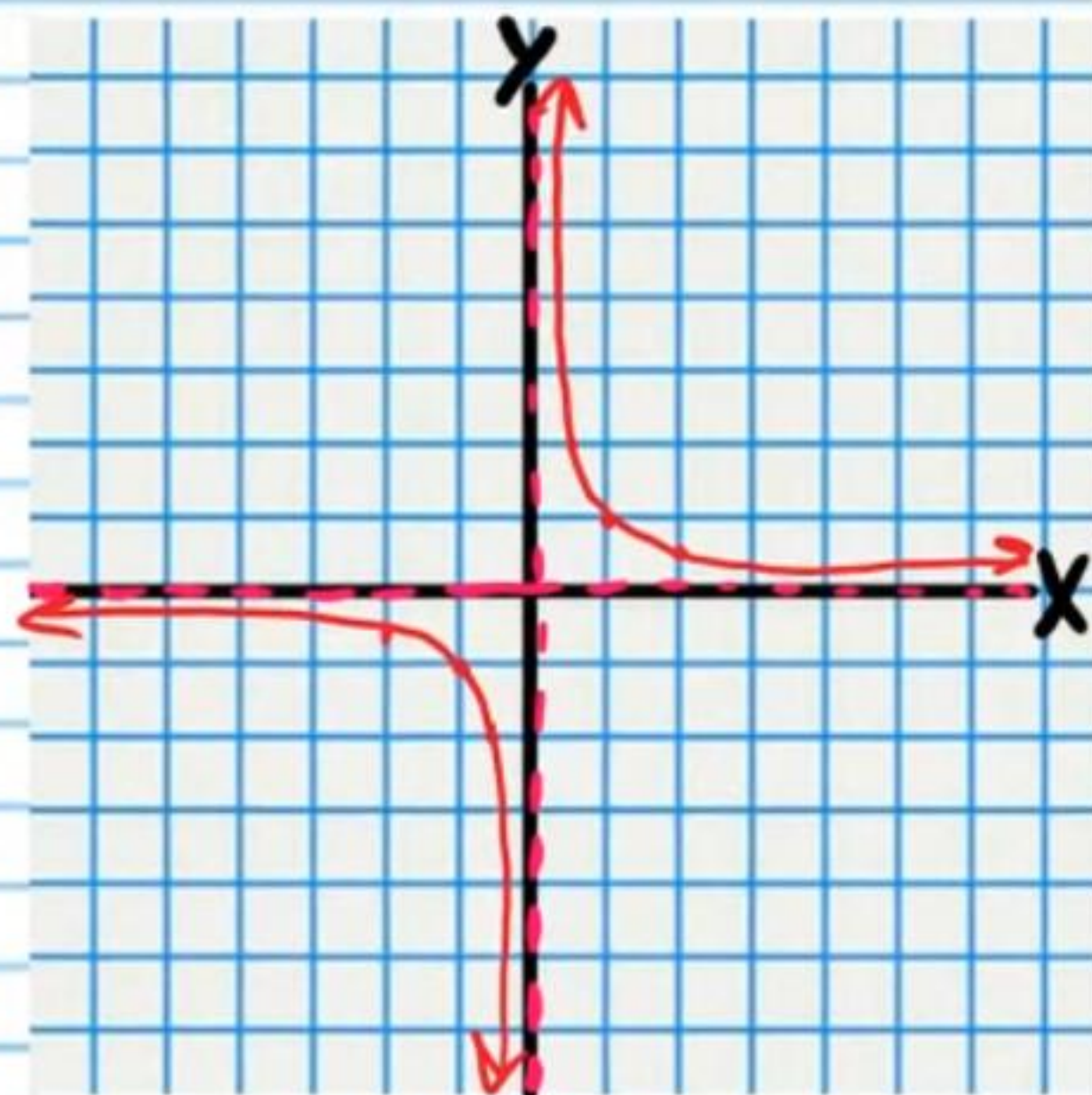
Parent Equation: $y = \frac{1}{x}$

$$y = -\frac{2}{x-2} - 2$$

$$y+2 = -\frac{2}{x-2}$$

$$\frac{y+2}{-2} = \frac{1}{x-2}$$

$$-\frac{(y+2)}{2} = \frac{1}{x-2}$$



24 Graph the equation above by performing multiple transformations on the parent graph. Draw the asymptote(s) as dotted lines.

Transformation #1

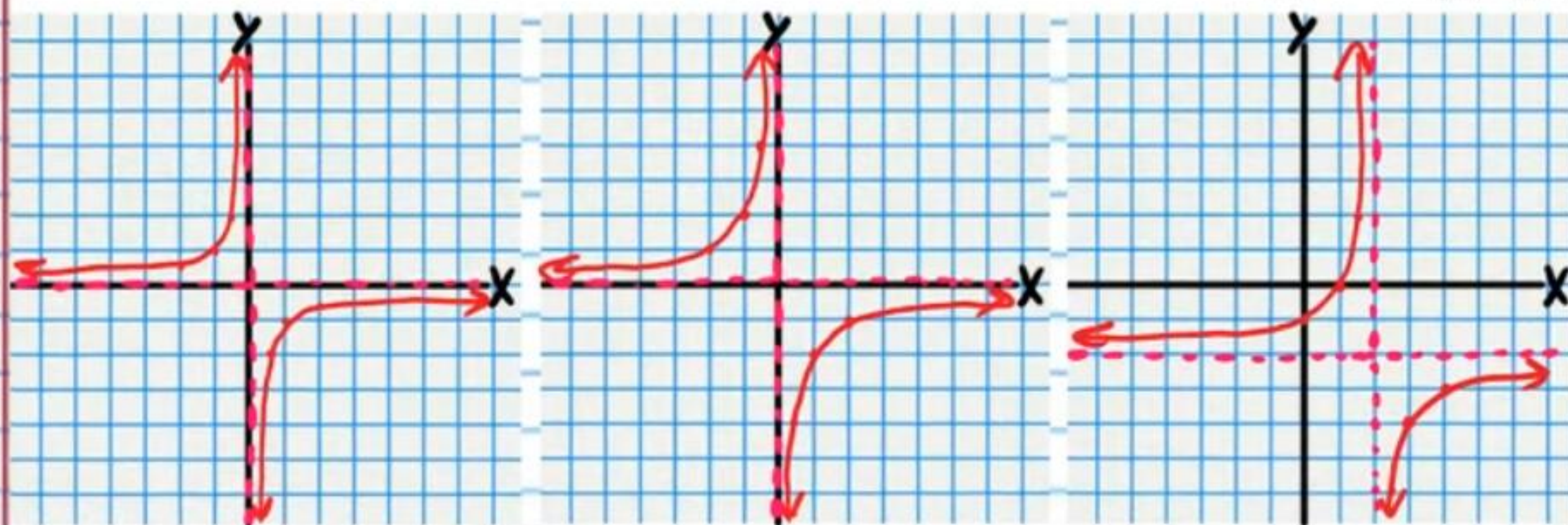
$$-y = \frac{1}{x}$$

Transformation #2

$$-\frac{y}{2} = \frac{1}{x}$$

Final Graph

$$-\frac{y+2}{2} = \frac{1}{x-2}$$



25 George has quarters, nickels, and dimes in his pocket, 30 coins in total. The number of quarters is one more than the number of dimes. How many of each type of coin does he have if the total value of the coins is \$4.20?

Q = # of quarters.

N = # of nickels

D = # of dimes

$$Q + N + D = 30$$

$$Q = 1 + D$$

$$0.25Q + 0.05N + 0.1D = 4.20$$

$$0.25(1+D) + 0.05N + 0.1D = 4.20$$

$$1 + D + N + D = 30$$

$$0.25 + 0.25D + 0.05N + 0.1D = 4.20$$

$$1 + 2D + N = 30$$

$$0.25 + 0.35D + 0.05N = 4.20$$

$$2D + N = 29$$

$$0.35D + 0.05N = 3.95$$

$$0.35D + 0.05N = 3.95$$

$$2D + N = 29$$

$$2(10) + N = 29$$

$$N = 9 \text{ nickels}$$

$$Q = 1 + 10$$

$$Q = 11 \text{ quarters}$$

$$\begin{array}{r} 0.35D + 0.05N = 3.95 \\ 0.1D + 0.05N = 1.45 \\ \hline \end{array}$$

$$0.25D = 2.5$$

$$D = 10 \text{ dimes}$$